claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a6d1d5680a65dc8a4.jsonl`
- SHA-256: `915944a68efb7a46d565c53f1f406442ae60a9badee84056cf04bdb26d01f573`
- Source modified: `2026-06-16T03:23:47+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:22:35.358Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"An ablation on THUMOS14 isolates the contribution of each component: a convolutional baseline pyramid scores 62.1% avg mAP; replacing it with the SGP layer raises this to 68.3% (+6.2), and adding the Trident-head on top yields 69.3% (a further +1.0 over the instant-level regression head) ? quantifying that the relative-boundary head adds a measurable, separable boundary-localization gain."

Source: https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf (primary)
Supporting quote: "compared with the baseline model we implement (Row 1), the SGP layer brings a 6.2% absolute improvement in the average mAP... we compare our Trident-head with the normal instant-level regression head"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:22:37.907Z)

I'll verify this claim against the primary source ? the TriDet CVPR 2023 paper. Let me fetch the ablation numbers directly.

3. assistant (2026-06-16T03:22:41.232Z)

I'll fetch the TriDet paper's ablation table directly to verify the specific numbers in the claim.

4. user (2026-06-16T03:22:45.046Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. assistant (2026-06-16T03:22:47.331Z)

The CVF server blocks the fetch. Let me try the arXiv version and search for the ablation numbers.

6. user (2026-06-16T03:22:51.232Z)

I cannot provide the specific ablation study details you're requesting based on the content provided. The webpage excerpt only contains the abstract and metadata for the TriDet paper from arXiv.

The abstract mentions that "TriDet hits an average mAP of 69.3% on THUMOS14," but it does not include the ablation study table with individual component performance metrics (baseline pyramid, SGP layer, and Trident-head contributions) or their exact mAP numbers.

To access this information, you would need to view the full PDF of the paper, which is available at the link provided on the arXiv page.

7. user (2026-06-16T03:22:57.634Z)

Web search results for query: "TriDet CVPR 2023 ablation THUMOS14 SGP layer 6.2 Trident-head average mAP baseline"

Links: [{"title":"TriDet - CVPR 2023 Open Access Repository","url":"https://openaccess.thecvf.com/content/CVPR2023/html/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.html"},{"title":"TriDet: Temporal Action Detection with Relative Boundary Modeling","url":"https://arxiv.org/pdf/2303.07347"},{"title":"TriDet: Temporal Action Detection","url":"https://www.emergentmind.com/papers/2303.07347"},{"title":"Noname manuscript No. (will be inserted by the editor)","url":"https://arxiv.org/pdf/2309.05590"},{"title":"[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling","url":"https://arxiv.org/abs/2303.07347"},{"title":"TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?","url":"https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"},{"title":"CVPR Poster TriDet: Temporal Action Detection With Relative Boundary Modeling","url":"https://cvpr.thecvf.com/virtual/2023/poster/21956"},{"title":"[Quick Review] TriDet: Temporal Action Detection with Relative Boundary Modeling","url":"https://liner.com/review/tridet-temporal-action-detection-with-relative-boundary-modeling"},{"title":"GitHub - dingfengshi/TriDet: [CVPR2023] Code for the paper, TriDet: Temporal Action Detection with Relative Boundary Modeling · GitHub","url":"https://github.com/dingfengshi/TriDet"}]

Based on the search results, I found the specific information you're looking for regarding the TriDet CVPR 2023 ablation study:

The SGP layer brings a 6.2% absolute improvement in the average mAP on THUMOS14 compared with the baseline model. The baseline model (without SGP and Trident-head components) achieved 62.1%

average mAP, while the full TriDet model with both SGP and Trident-head achieved 69.3% average mAP.

The ablation study breaks down the effectiveness of the individual components:
- The study demonstrates the effectiveness of the proposed components in TriDet: SGP layer and Trident-head

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:23:00.666Z)

The arXiv PDF is the authoritative source. Let me fetch the full PDF text to verify the exact ablation table rows, especially the 62.1 ? 68.3 ? 69.3 progression and the "+1.0 over instant-level regression head" detail.

9. user (2026-06-16T03:23:07.639Z)

I've carefully reviewed the provided PDF content, which appears to be a research paper with figures and embedded fonts. Unfortunately, the document is heavily compressed and encoded (FlateDecode format), making the actual table data illegible in the raw content provided.

While I can see references to objects related to figures (framework.pdf, cosine.pdf, intro.pdf) and font definitions, the actual ablation study table with specific mAP numbers for the baseline, SGP layer, and Trident-head configurations is not readable in this encoded format.

To provide the exact metrics you're requesting?the baseline mAP, the ~6.2% improvement from SGP, and the specific comparison with the normal regression head?I would need either:
1. The paper in plain text or standard PDF format
2. A screenshot or image of the actual table
3. The rendered text content rather than the encoded stream data

I cannot provide the verbatim quotes and exact table numbers without access to the readable content.

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579910495-xf7r2b.pdf]

10. assistant (2026-06-16T03:23:09.283Z)

The PDF was saved locally. Let me read it directly to extract the exact ablation table.

11. user (2026-06-16T03:23:11.925Z)

pdfplumber available
pymupdf available
pypdf available

12. user (2026-06-16T03:23:18.101Z)

62.1 -> 1 hits
68.3 -> 8 hits
69.3 -> 9 hits
6.2 -> 10 hits
SGP -> 45 hits
Trident-head -> 33 hits
instant-level -> 5 hits
baseline -> 14 hits

13. assistant (2026-06-16T03:23:21.920Z)

All key numbers are present. Notably 62.1 appears only once. Let me extract the context around the ablation table and the relevant numbers.

14. user (2026-06-16T03:23:25.946Z)

==== context for '62.1' at 29005 ====
tional layers is set to 3 and the number of channels of the intermediate feature is set to the same dimension as the Table 5. Analysis of the Effectiveness of three main components on THUMOS14. Method SA SGP Trident 0.3 0.5 0.7 Avg. 1 77.3 65.2 40.0 62.1 2 ? 82.1 71.0 43.9 66.8 3 ? 83.6 71.7 45.8 68.3 4 ? ? 83.6 72.9 47.4 69.3 Table 6. Analysis of computation cost on THUMOS14. Main: All parts of the model except the detection head. *: Our method with a normal instant-level regression head. Method mAP GMACs Latency (ms) 0.3 0.7 Avg. Main Head All ActionFormer 82.1 43.9 66.8 30.8 14.4 45.3 224 TriDet? 83.6 45.8 68.3 14.5 14.4 28.9 145 TriDet 83

==== context for 'the SGP layer brings' at 29673 ====
29.1 43.7 167 intermediate dimension in the FFN in our SGP layer. All other hyperparameters (e.g. number of the pyramid layers, etc.) are set to the same as our framework. As depicted in Tab. 5, compared with the baseline model we implement (Row 1), the SGP layer brings a 6.2% absolute improvement in the average mAP. Secondly, we com- pare the SGP with the previous state-of-the-art method, Ac- tionFormer, which adopts a self-attention mechanism in a sliding window behavior [4] with window size 7 (Row 2). We can see our SGP layer still has 1.5% improvement in average mAP, demonstrating that the convolutional network can also have excellent p

==== context for 'instant-level regression head' at 29219 ====
t 0.3 0.5 0.7 Avg. 1 77.3 65.2 40.0 62.1 2 ? 82.1 71.0 43.9 66.8 3 ? 83.6 71.7 45.8 68.3 4 ? ? 83.6 72.9 47.4 69.3 Table 6. Analysis of computation cost on THUMOS14. Main: All parts of the model except the detection head. *: Our method with a normal instant-level regression head. Method mAP GMACs Latency (ms) 0.3 0.7 Avg. Main Head All ActionFormer 82.1 43.9 66.8 30.8 14.4 45.3 224 TriDet? 83.6 45.8 68.3 14.5 14.4 28.9 145 TriDet 83.6 47.4 69.3 14.5 29.1 43.7 167 intermediate dimension in the FFN in our SGP layer. All other hyperparameters (e.g. number of the pyramid layers, etc.) are set to the same as our framework. As depicted in Tab. 5, c

15. assistant (2026-06-16T03:23:36.213Z)

I now have the exact ablation table (Table 5) from the primary source. Let me carefully cross-check the claim against the actual data.

Table 5 (Effectiveness of three main components on THUMOS14):

Row	SA	SGP	Trident	0.3	0.5	0.7	Avg.
1				77.3	65.2	40.0	**62.1**
2	? (SA)			82.1	71.0	43.9	66.8
3		? (SGP)		83.6	71.7	45.8	**68.3**
4		?	?	83.6	72.9	47.4	**69.3**

Now I'll verify each component of the claim against this:

1. **"convolutional baseline pyramid scores 62.1% avg mAP"** ? Row 1 = 62.1. ? Correct. The text confirms "the baseline model we implement (Row 1)."

2. **"replacing it with the SGP layer raises this to 68.3% (+6.2)"** ? Row 3 (SGP) = 68.3, and 68.3 ? 62.1 = 6.2. ? The paper text explicitly states: "the SGP layer brings a 6.2% absolute improvement in the average mAP."

3. **"adding the Trident-head on top yields 69.3% (a further +1.0 over the instant-level regression head)"** ? Row 4 (SGP + Trident) = 69.3, and 69.3 ? 68.3 = 1.0. ? Row 3 (SGP, no Trident) uses the instant-level regression head. Table 6 confirms this: "TriDet?" (with "a normal instant-level regression head") = 68.3 Avg, and full TriDet = 69.3 Avg. So the Trident-head adds exactly +1.0 over the instant-level regression head.

4. **"quantifying that the relative-boundary head adds a measurable, separable boundary-localization gain"** ? Reasonable interpretation. The gain is most pronounced at the stricter tIoU=0.7 (45.8 ? 47.4, +1.6), consistent with the Trident-head improving boundary localization. This is a fair characterization, not an overreach.

Every number checks out exactly against the primary source. The claim accurately reconstructs the ablation, including the subtle "+1.0 over the instant-level regression head" detail (corroborated independently by both Table 5 Row 3?4 and Table 6's TriDet? row). Source is the CVPR 2023 primary paper, current (2023), not marketing.

16. user (2026-06-16T03:23:44.463Z)

Structured output provided successfully

17. assistant (2026-06-16T03:23:47.034Z)

Verified against the primary source. The claim's numbers (62.1 ? 68.3 ? 69.3, +6.2 for SGP, +1.0 for Trident-head over the instant-level regression head) all reproduce exactly from TriDet's Table 5 and Table 6. refuted=false, high confidence.